

Ethanol Blending in Gasoline - Ecuador

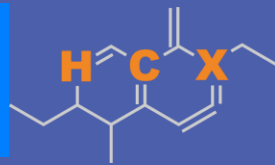
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**U.S. GRAINS &
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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



Ethanol Blending in Gasoline - Ecuador

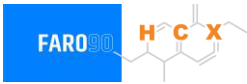


In 2024, gasoline consumption in Ecuador was 1,188 million gallons (4,498 million liters). A new plan was established to implement new grades of gasoline: four grades with 5 different types according to the current quality norm (NTE INEN 935:2021): Extra (RON 85), Ecopais E10 (RON 85), Ecoplus (RON 89), Súper (RON 92) and Super Premium (RON 95). Super Premium (RON 95) will gradually substitute Super (RON 92). Gasoline Ecoplus (RON 89) was recently introduced in the market. Ecuador has local production and imports naphtha (unfinished gasolines) to supply national demand.

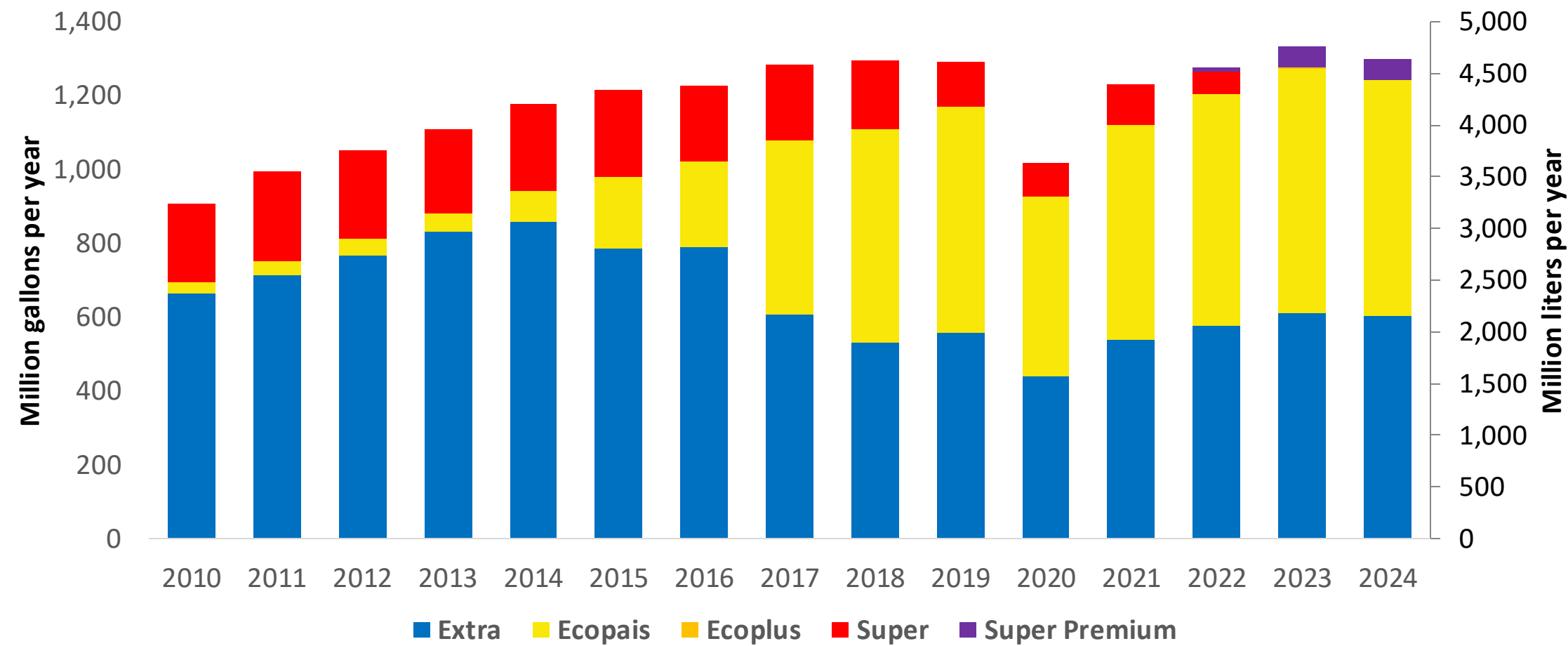
There is no specific mandate for ethanol blending, however there is a permit to blend it with gasoline. Ethanol has been blending in gasoline since 2009. Today the usual blend is E10 for Ecoplus and Super. Ecopais gasoline is supplied in some provinces with an ethanol content between 3-4%. Gasoline Ethanol is produced, consumed and exported to neighboring countries for fuel and industrial use.

Source: Ministerio de Energía y Minas, EP PETROECUADOR, 2024

Gasoline Demand in Ecuador

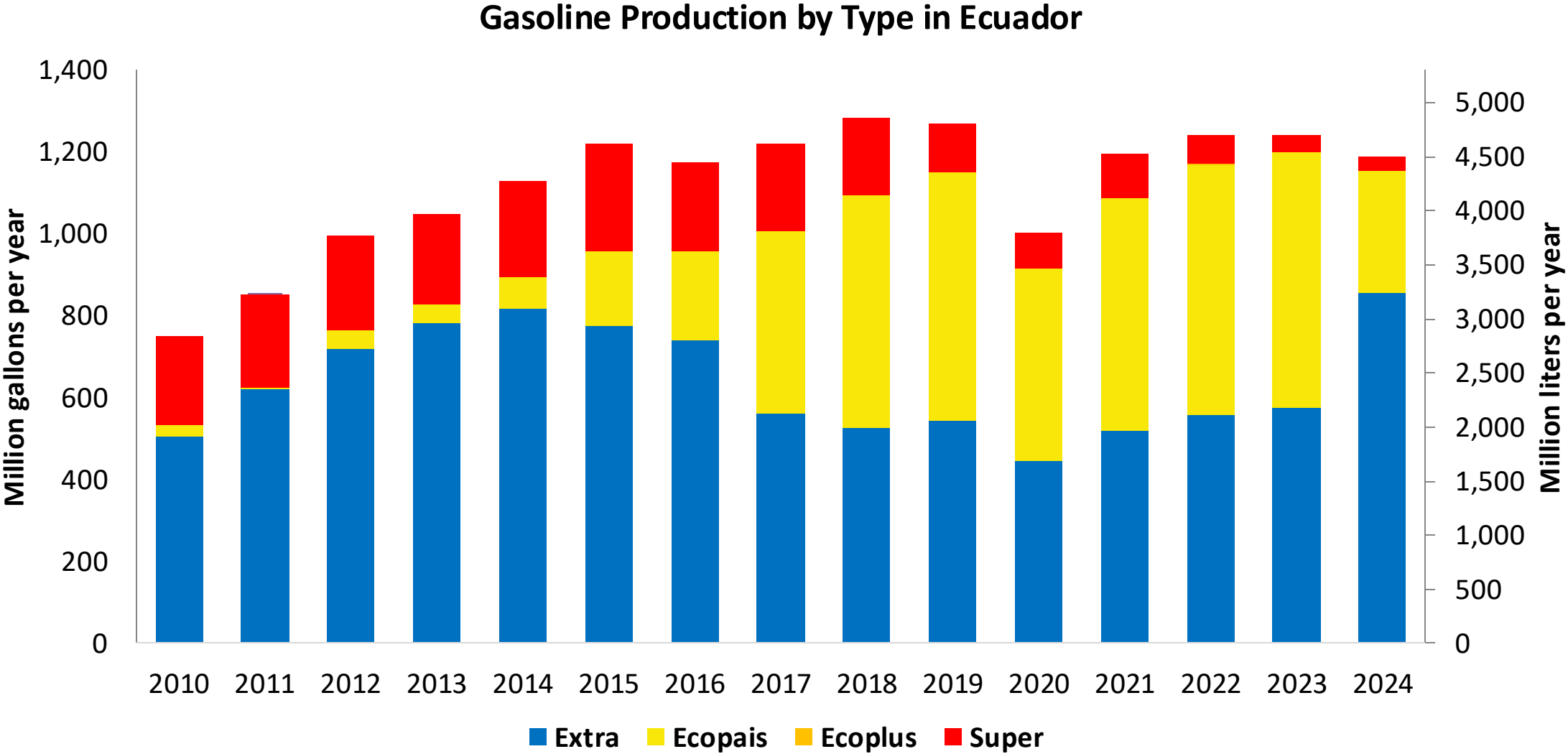


Gasoline Demand by Type in Ecuador



Source: EP PETROECUADOR, 2024

Gasoline Production in Ecuador

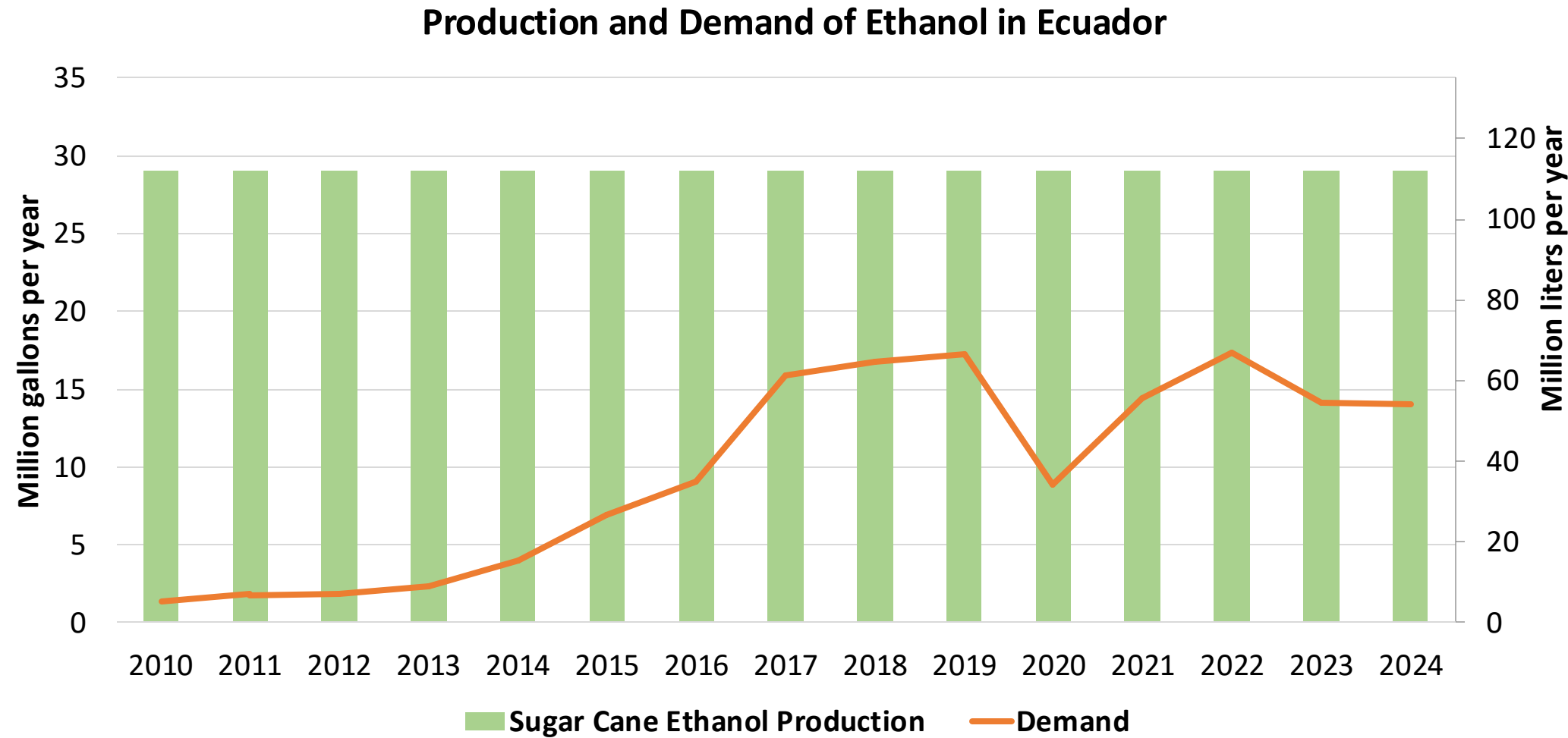
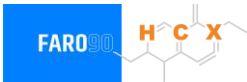


Source: EP PETROEQUADOR, 2024

The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexene ring. The structure features a double bond between carbons 1 and 2, with substituents at various positions: a methyl group at C3, a methyl group at C4, a methyl group at C5, and a methyl group at C6. The carbons are labeled with orange letters: H at C1, C at C2, X at C3, and a methyl group at C4. The FARO90 logo is a blue square with the text 'FARO90' in white.



Ethanol Balance in Ecuador



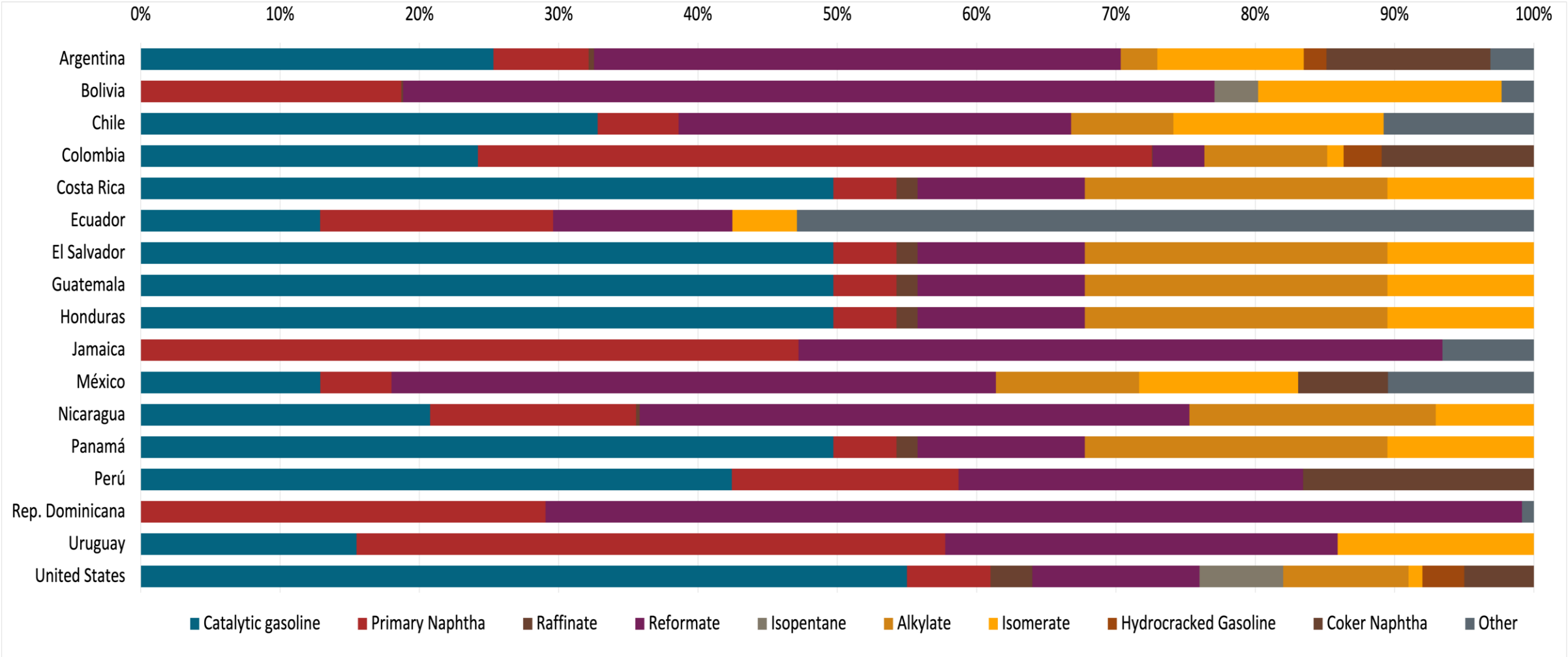
Source: EP PETROECUADOR, 2024; CEDRSSA, 2020

Gasoline Quality in Ecuador

Name	NTE INEN 935:2021				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021				2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	RON 85	RON 89	RON 92	RON 93	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 2 %v/v	<2 %v/v	<1,3 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 30 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 18 %v/v	< 25 %v/v	< 25 %v/v	< 25 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0 mg/l	< 0 mg/l	< 0 mg/l	< 0 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	0 mg/l	0 mg/l	0 mg/l	0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	85	89	92	95	> 95	> 95	> 98	> 98
MON					> 85	> 88	> 85	> 88
AKI								
Sulfur Content	650 mg/kg	650 mg/kg	450 mg/kg	<300 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)					<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<60 kPa	<60 kPa	<60 kPa	<62 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



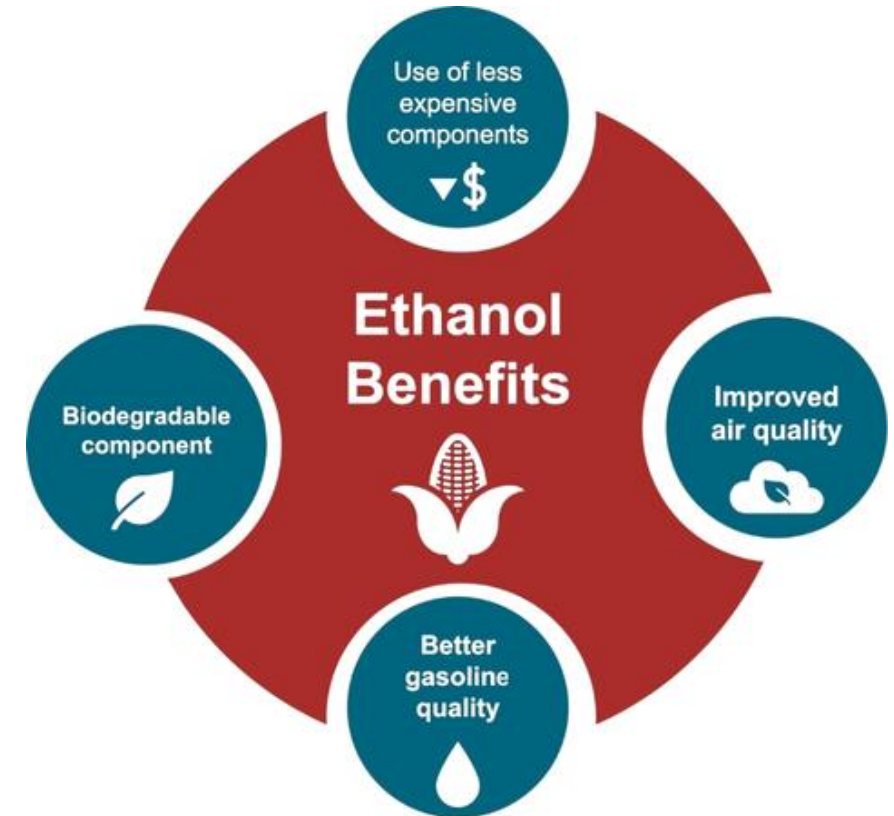
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

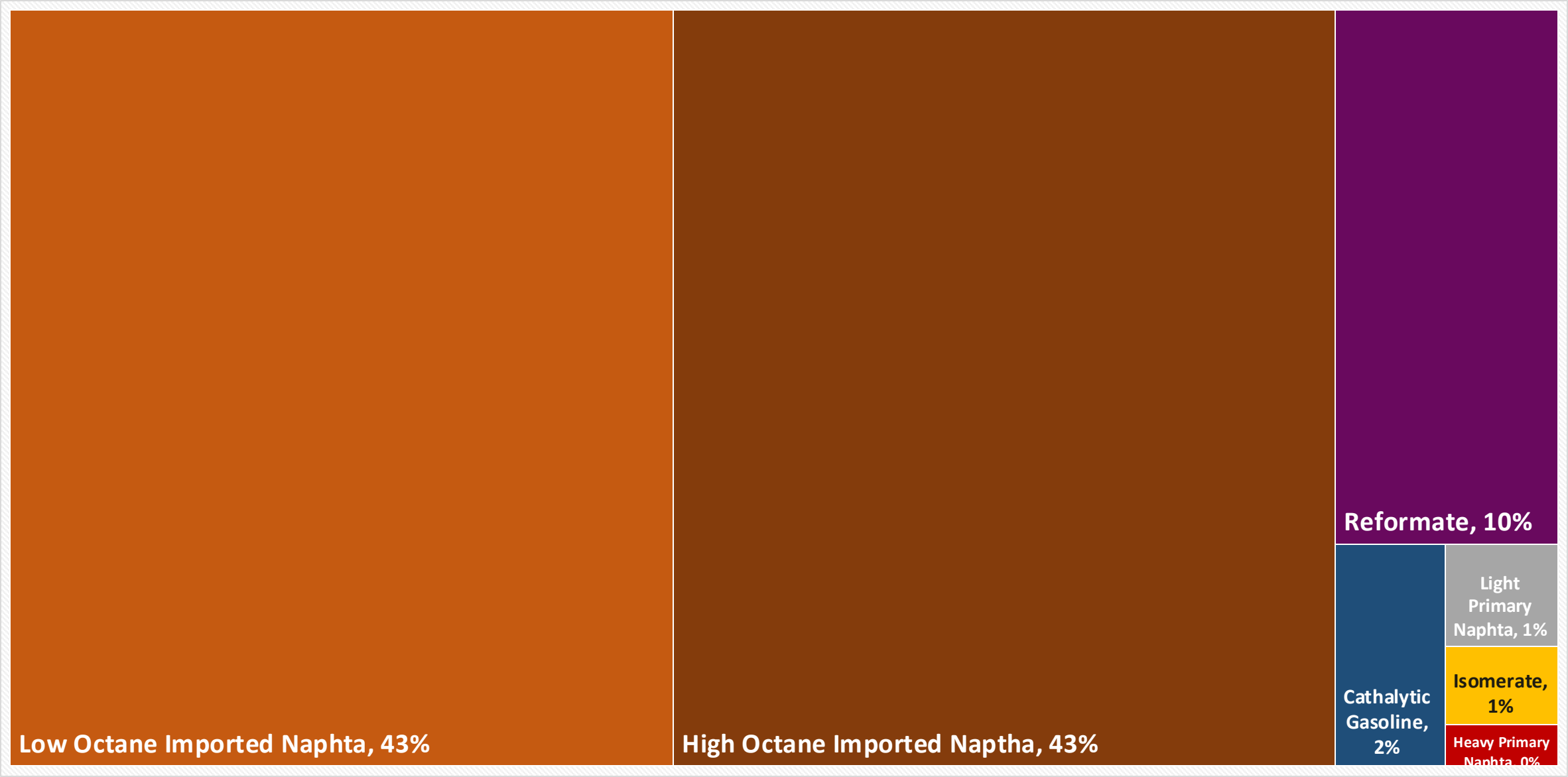
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

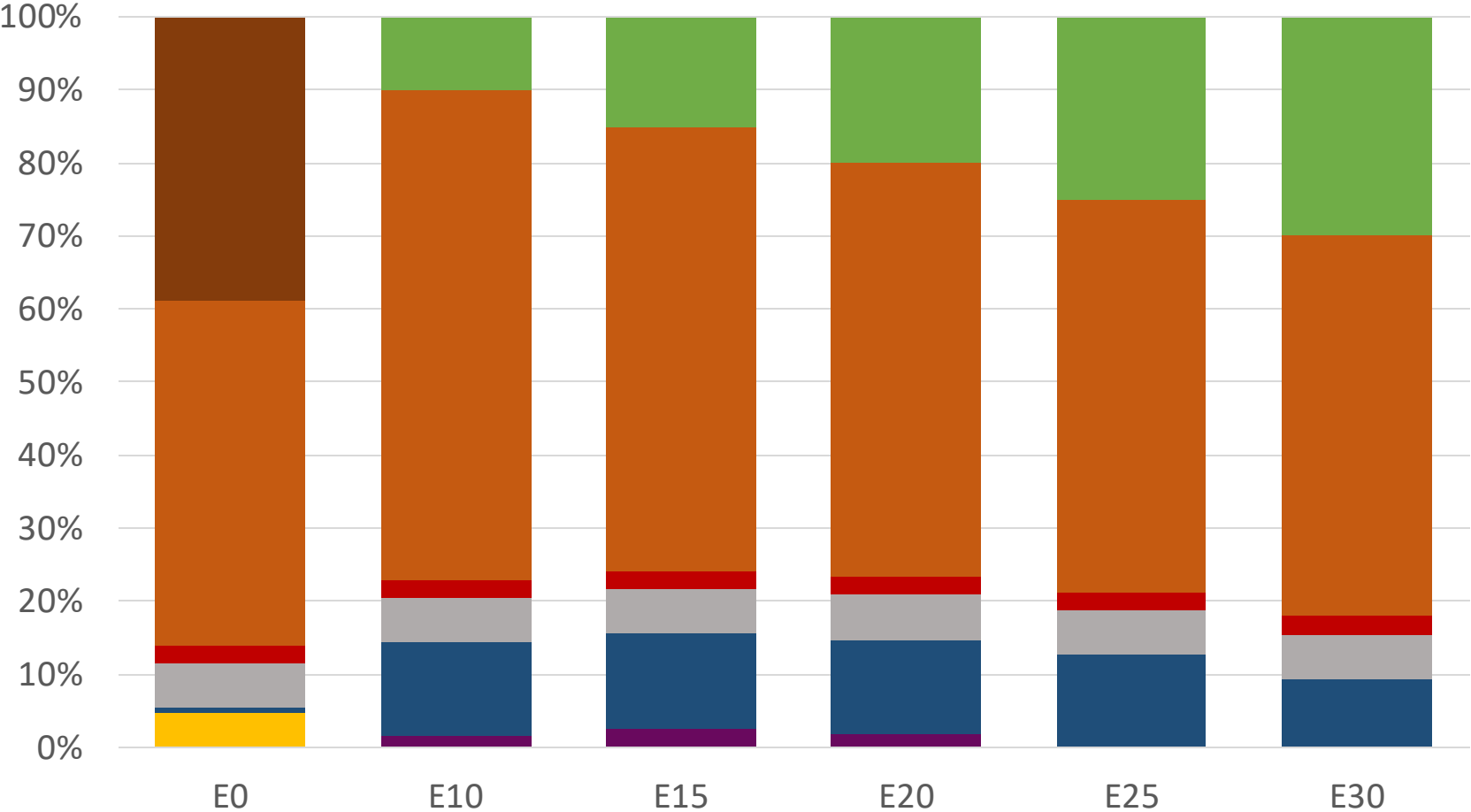
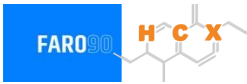
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Available Blending Components

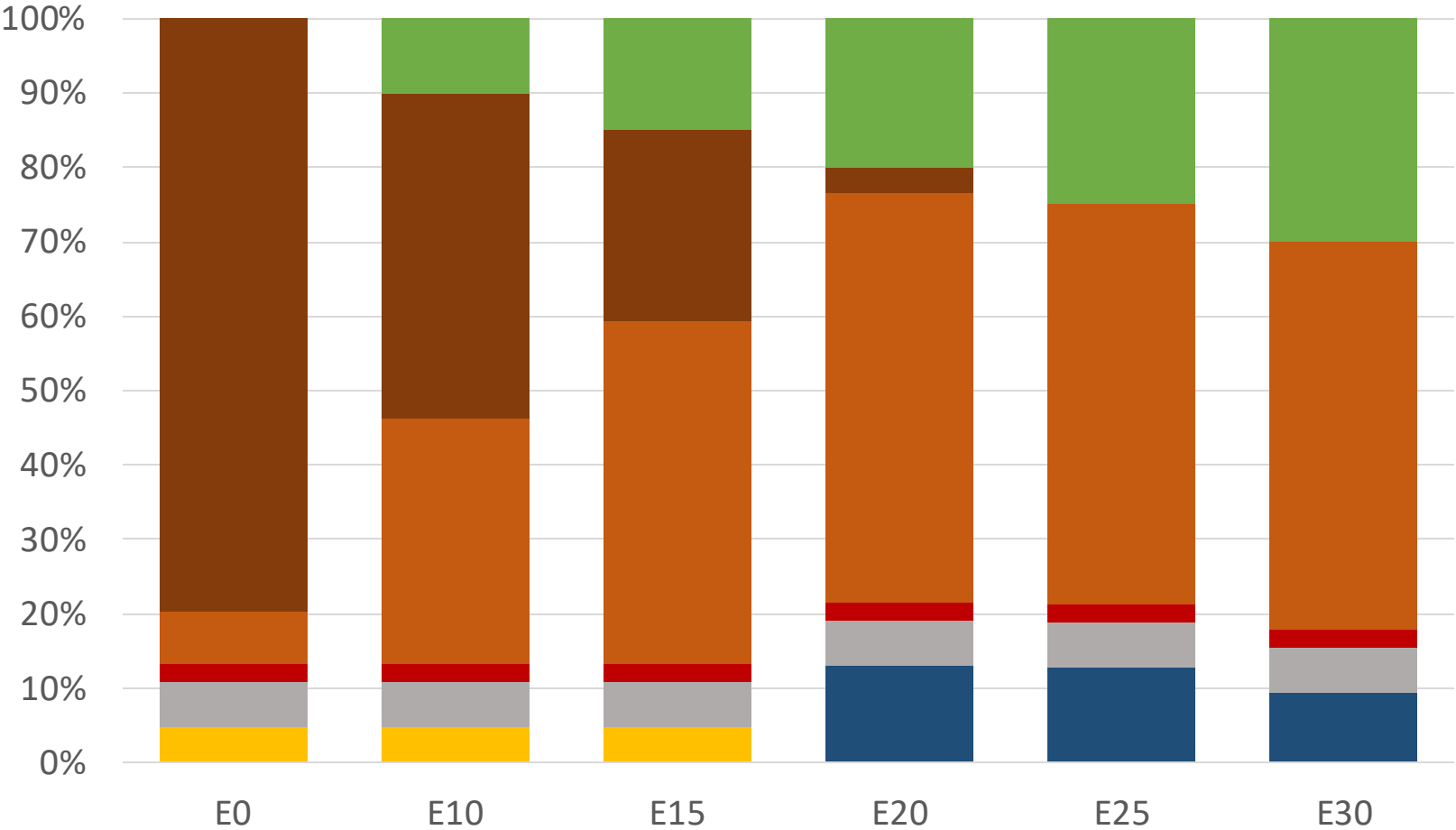


Ecuador –RON 85 Gasoline – Constant Octane



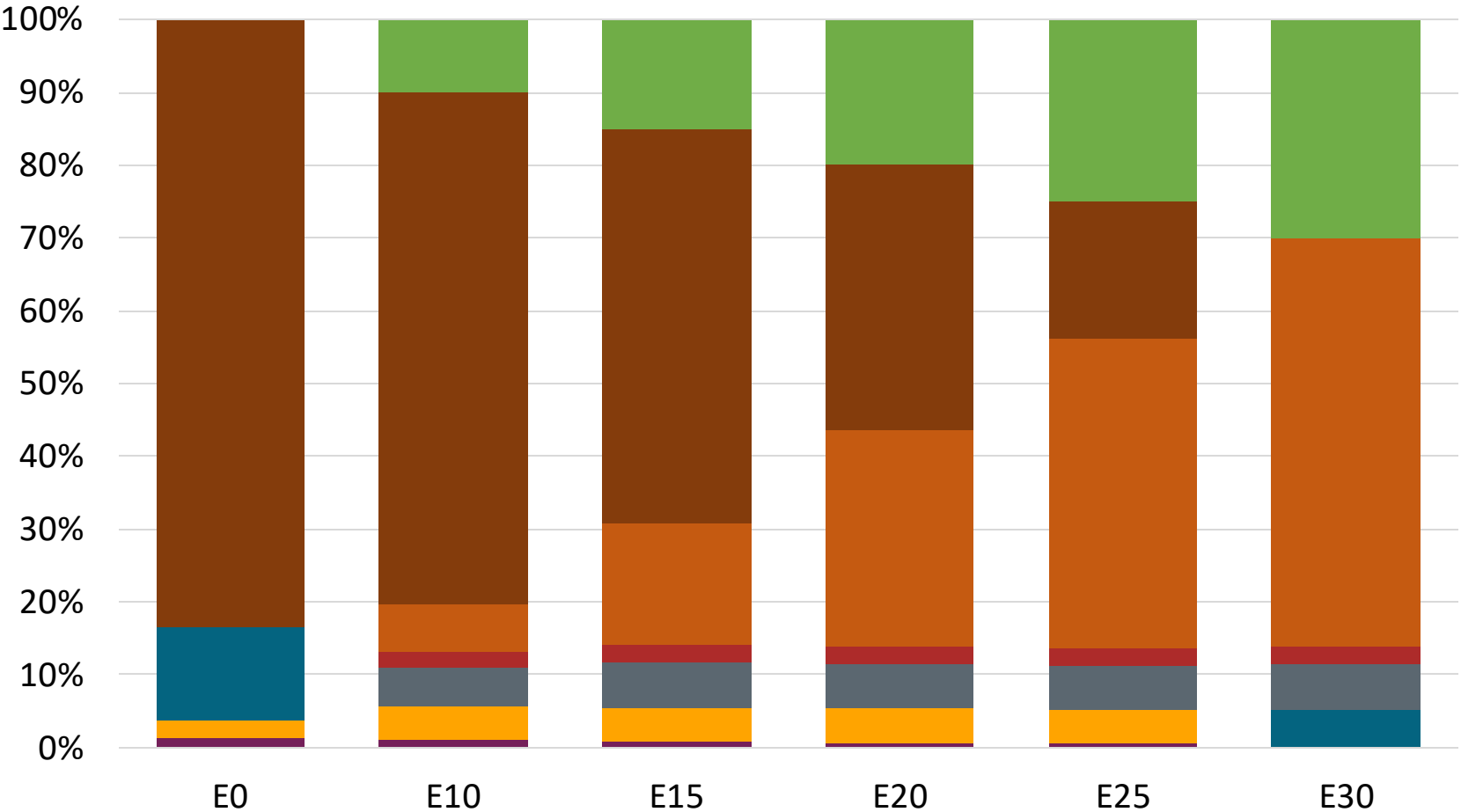
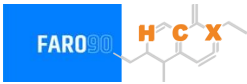
Octane (RON)	85.0	85.9	87.9	89.6	91.0	92.5
Price (US\$/gal)	\$ 2.386	\$ 2.321	\$ 2.310	\$ 2.288	\$ 2.258	\$ 2.234

Ecuador – RON 89 Gasoline– Constant Octane



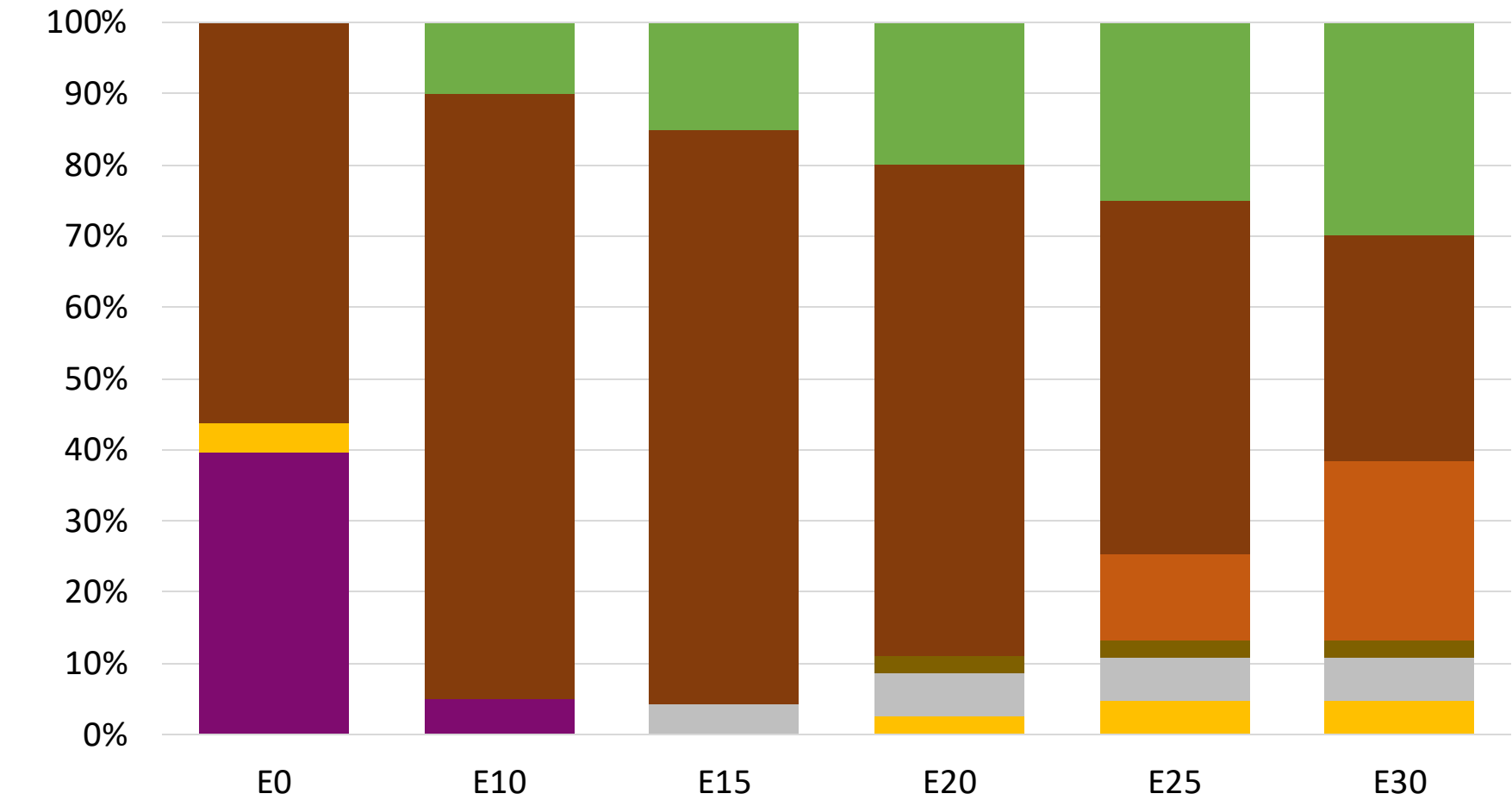
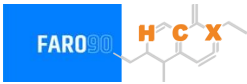
Octane (RON)	89.0	89.0	89.0	89.6	91.0	92.5
Price (US\$/gal)	\$ 2.453	\$ 2.357	\$ 2.310	\$ 2.282	\$ 2.258	\$ 2.234

Ecuador – RON 92 Gasoline – Constant Octane



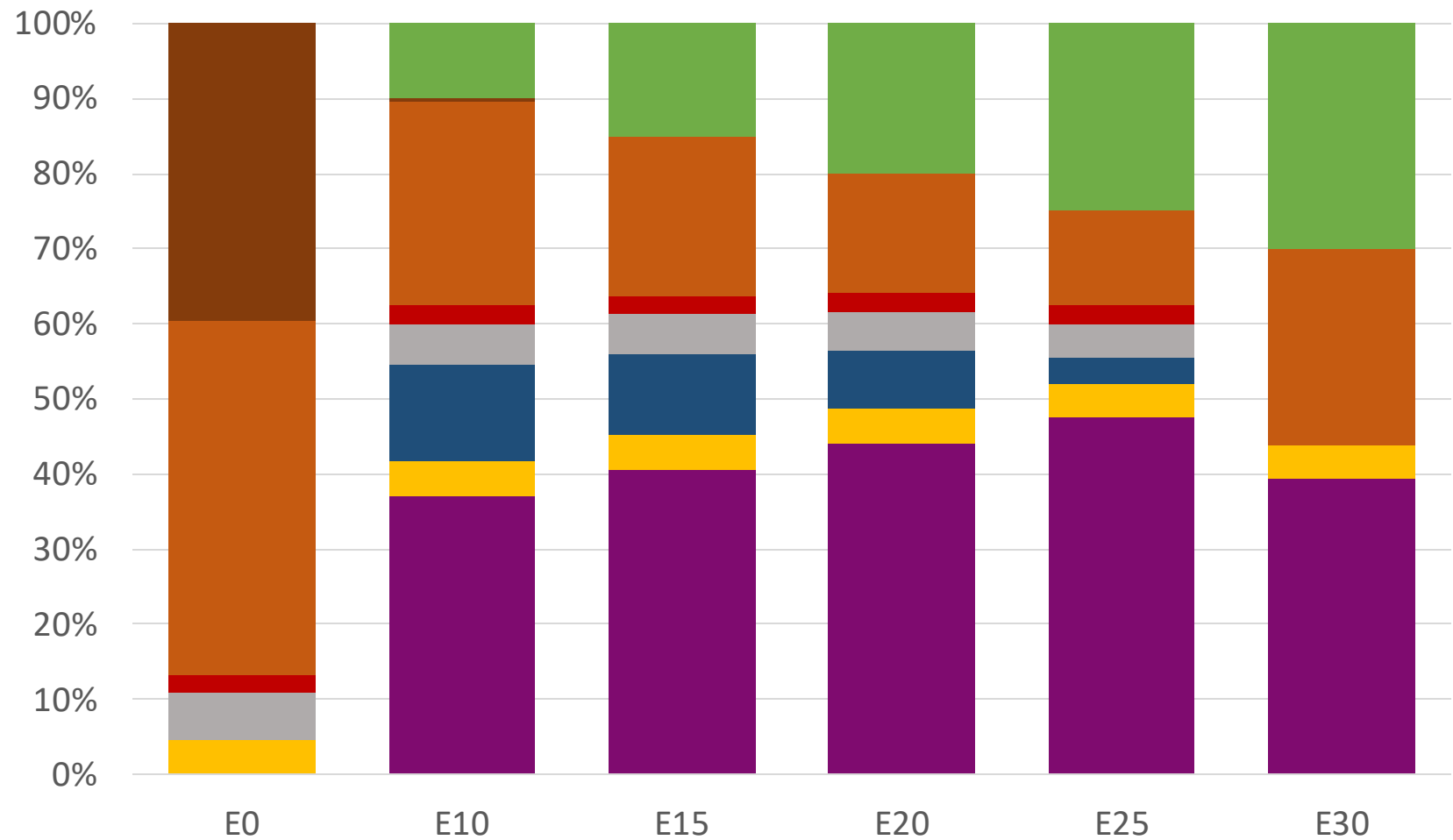
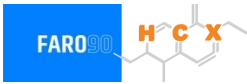
Octane (RON)	92.0	92.0	92.0	92.0	92.0	92.1
Price (US\$/gal)	\$ 2.584	\$ 2.419	\$ 2.362	\$ 2.314	\$ 2.266	\$ 2.225

Ecuador – RON 95 Gasoline – Constant Octane



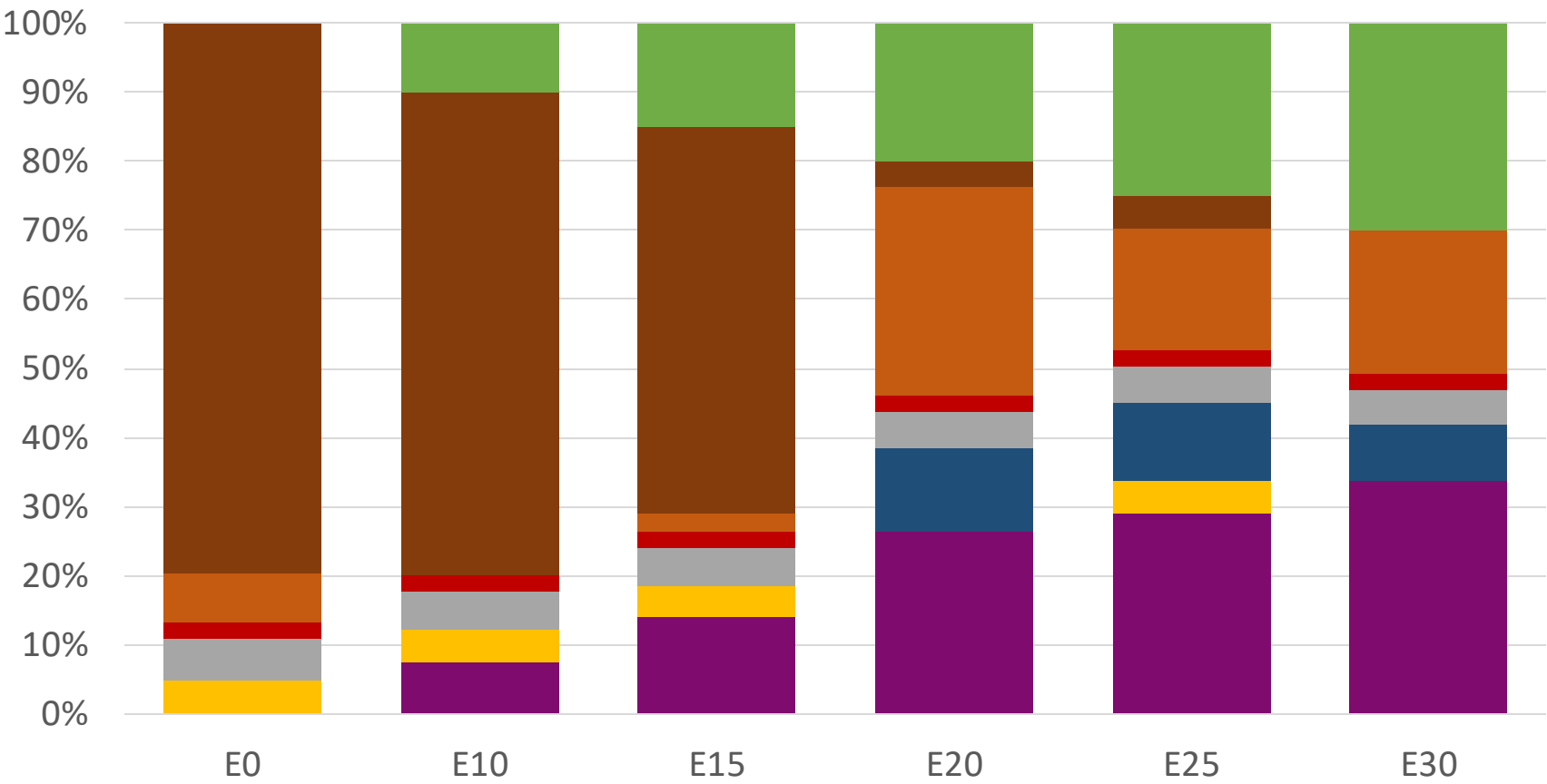
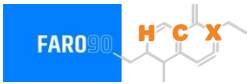
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US\$/gal)	\$ 2.748	\$ 2.548	\$ 2.454	\$ 2.364	\$ 2.315	\$ 2.267

Ecuador – RON 85 Gasoline – Increased Octane



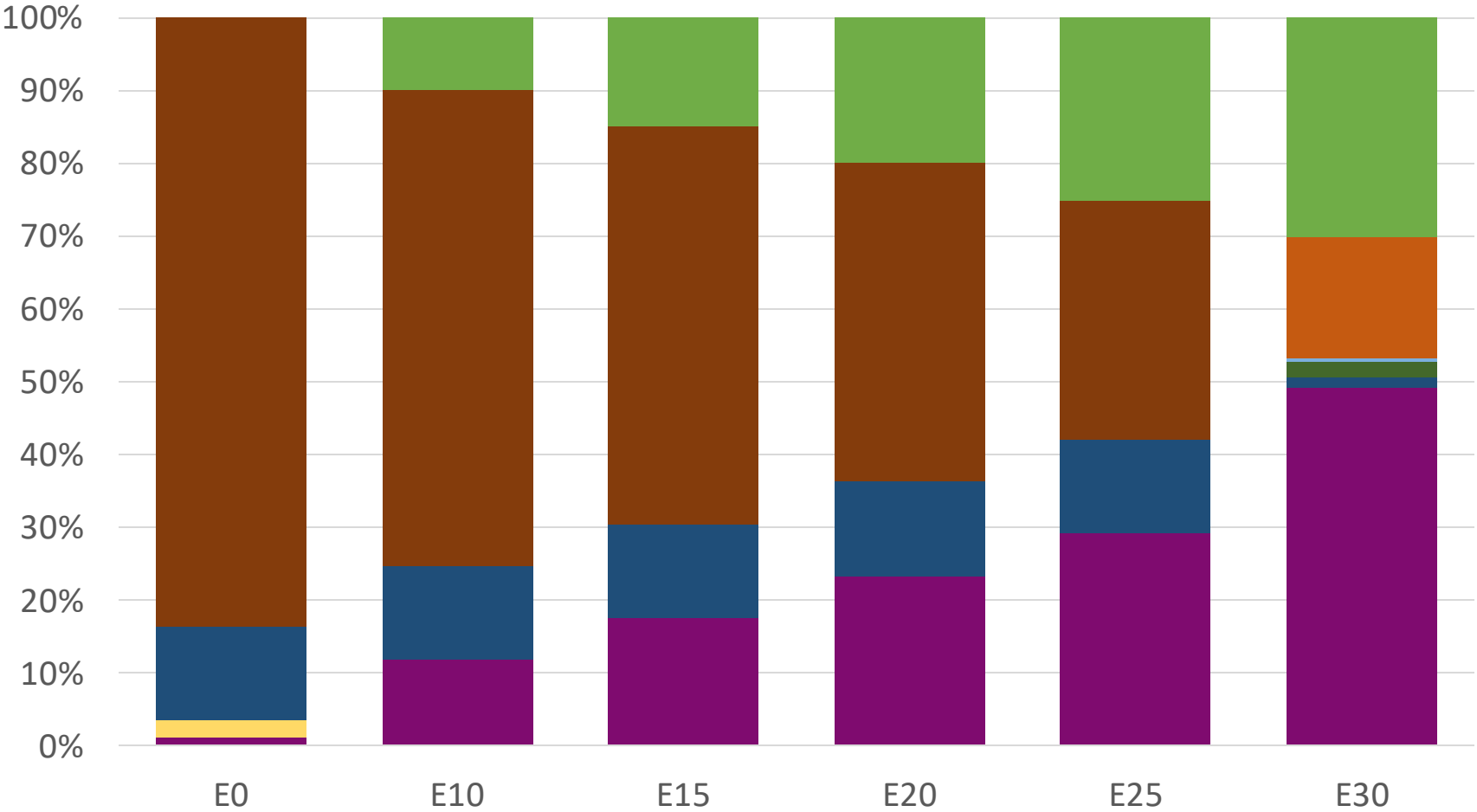
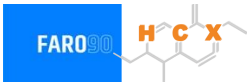
Octane (RON)	85.0	92.7	94.9	97.1	99.1	100.1
Price (US\$/gal)	\$ 2.386	\$ 2.386	\$ 2.386	\$ 2.386	\$ 2.386	\$ 2.386

Ecuador – Gasoline RON 89 – Increased Octane



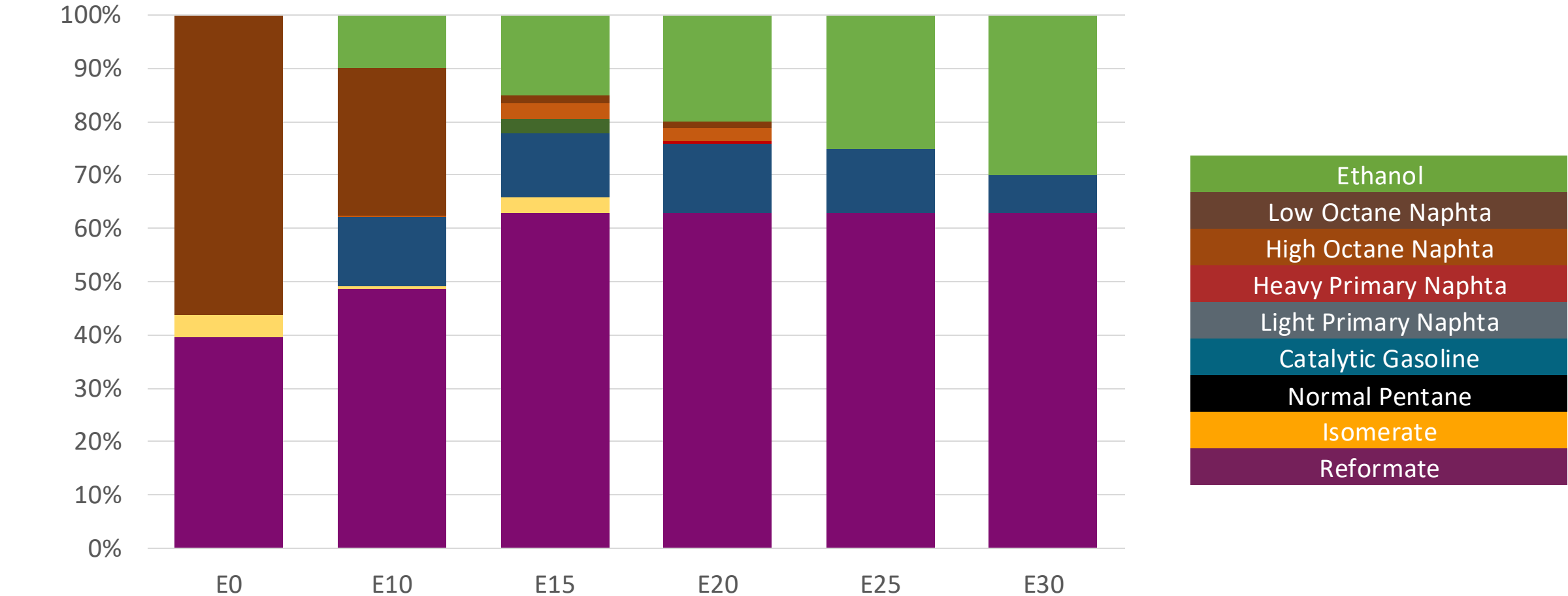
Octane (RON)	89.0	93.0	94.6	94.4	97.0	98.6
Price (US\$/gal)	\$ 2.453	\$ 2.453	\$ 2.453	\$ 2.453	\$ 2.453	\$ 2.453

Ecuador – RON 92 Gasoline – Increased Octane



Octane (RON)	92.0	95.5	97.3	99.1	100.8	101.4
Price (US\$/gal)	\$ 2.584	\$ 2.584	\$ 2.584	\$ 2.584	\$ 2.584	\$ 2.584

Ecuador – RON 95 Gasoline – increased Octane



Octane (RON)	95.0	98.4	99.9	101.8	103.5	104.8
Price (US\$/gal)	\$ 2.748	\$ 2.748	\$ 2.748	\$ 2.748	\$ 2.748	\$ 2.748

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

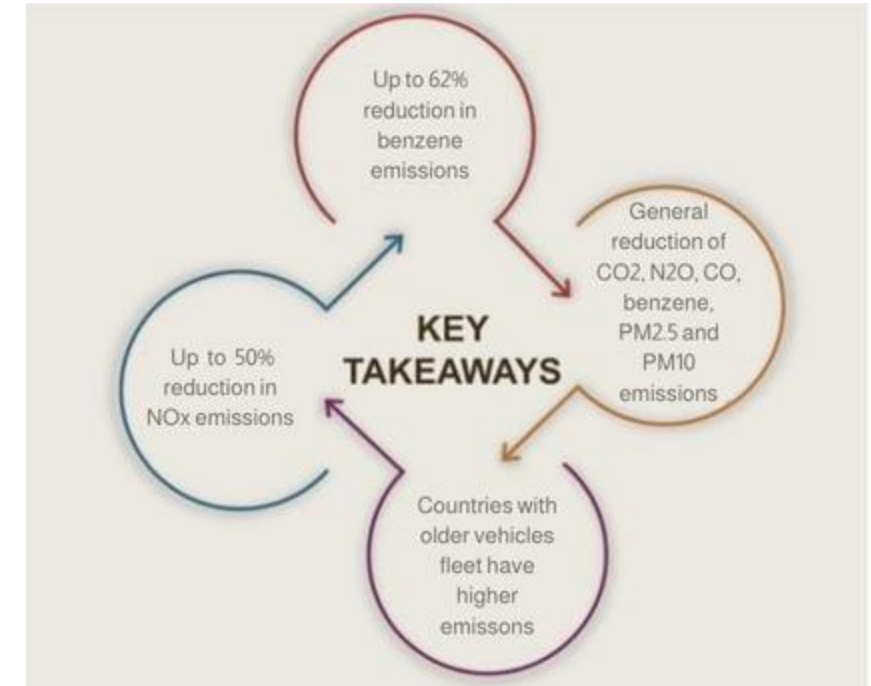
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

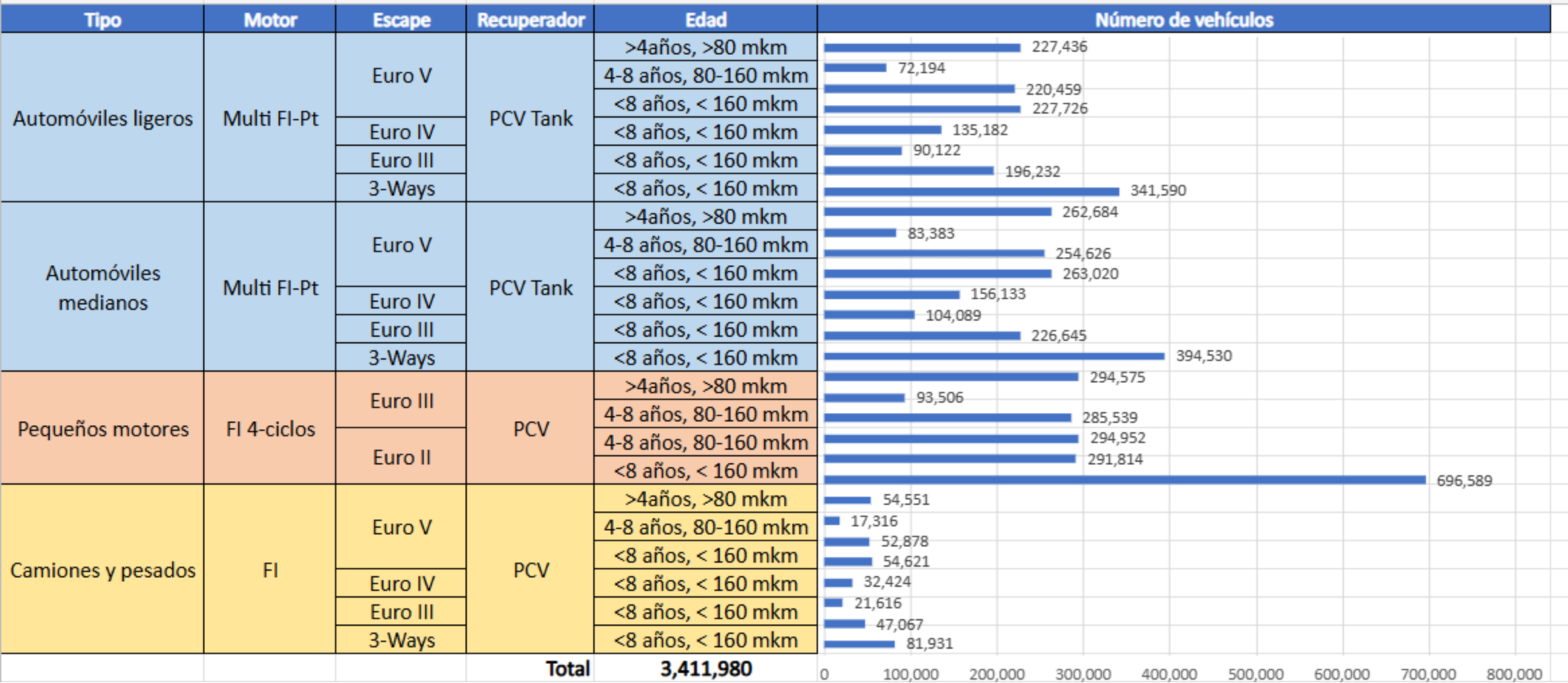
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Ecuador



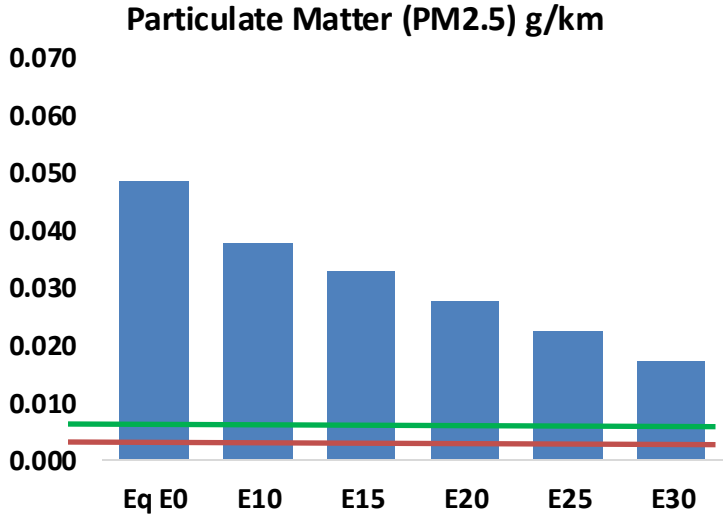
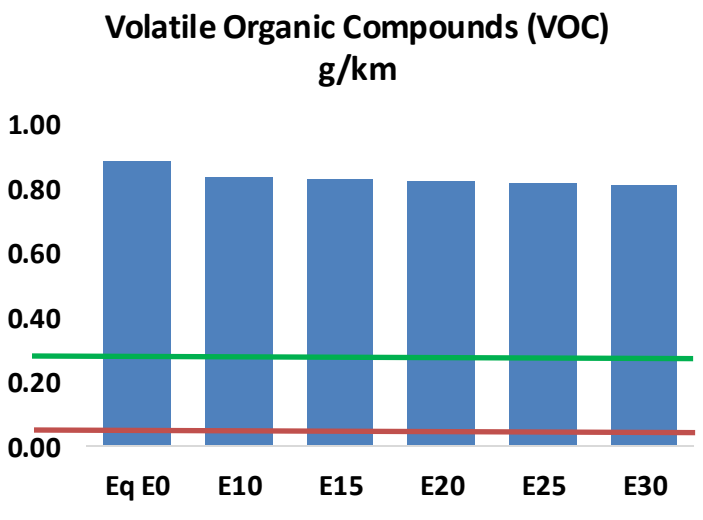
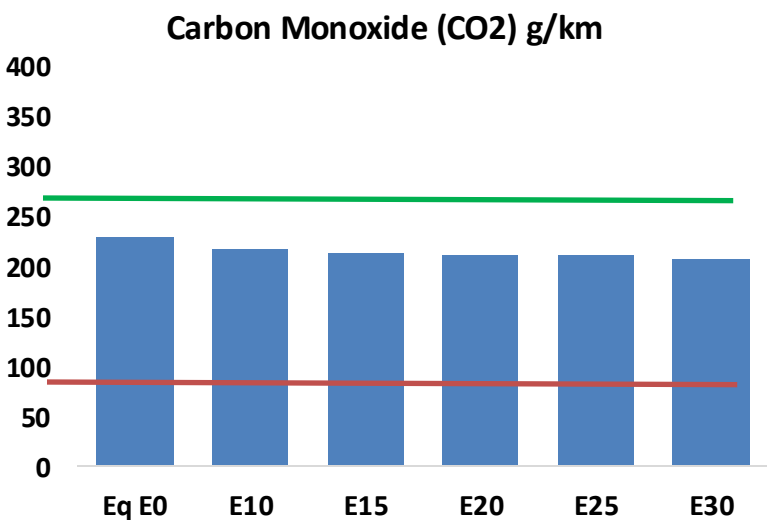
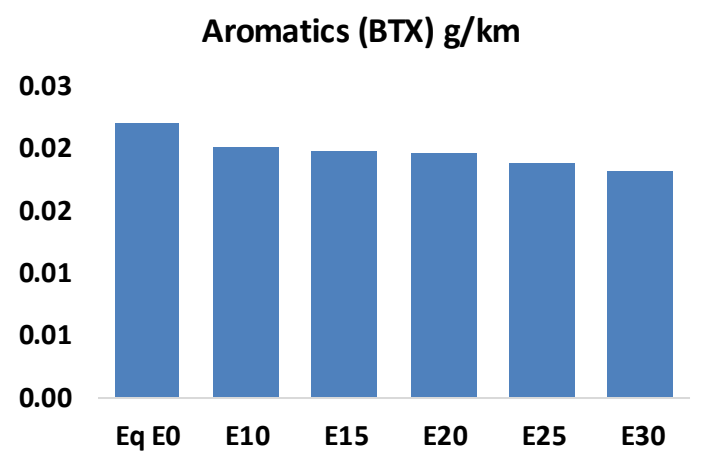
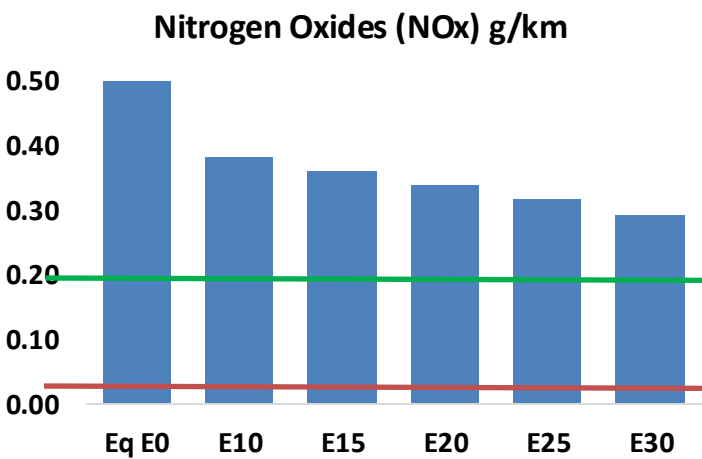
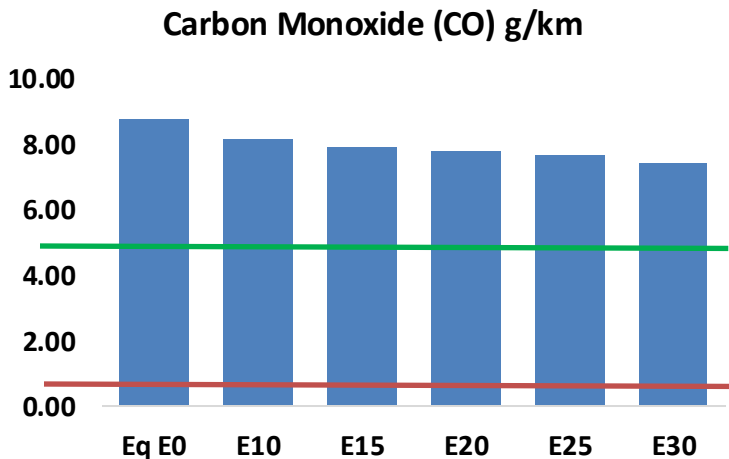
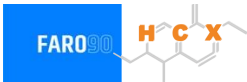
Vehicle Fleet: **3,411,980**

Age: 13 years

Source: CINAIE, 2023; analysis de Faro90

Ecuador – Vehicular Emissions

United States
Euro 6



Ecuador – Gasoline Vehicle Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	846,294	814,835	783,377	764,154	747,828	736,184	717,432	-7%	-12%
CO2	22,073,764	21,506,153	20,970,076	20,548,467	20,339,838	20,243,798	19,870,646	-5%	-8%
VOC	85,060	82,861	80,663	79,709	79,138	78,817	77,973	-5%	-7%
NOx	52,385	39,288	36,669	34,561	32,594	30,425	28,131	-30%	-38%
SOx	277	256	235	218	201	182	163	-15%	-28%
NH3	6,269	6,207	6,145	6,174	6,243	6,292	6,333	-2%	0%
N2O	829	824	821	825	842	851	861	-1%	2%
CH4	19,548	19,548	19,548	19,841	20,271	20,643	20,995	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	694	676	659	651	646	644	637	-5%	-7%
Acetaldehyde	1,417	1,626	2,386	3,633	4,950	5,656	6,683	68%	249%
Formaldehyde	5,279	5,132	5,983	7,025	7,360	8,142	8,864	13%	39%
Benzene	2,120	2,038	1,931	1,900	1,884	1,802	1,744	-9%	-11%
PM2.5	1,564	1,388	1,213	1,053	895	726	550	-22%	-43%
PM10	3,103	2,756	2,408	2,091	1,776	1,441	1,092	-22%	-43%

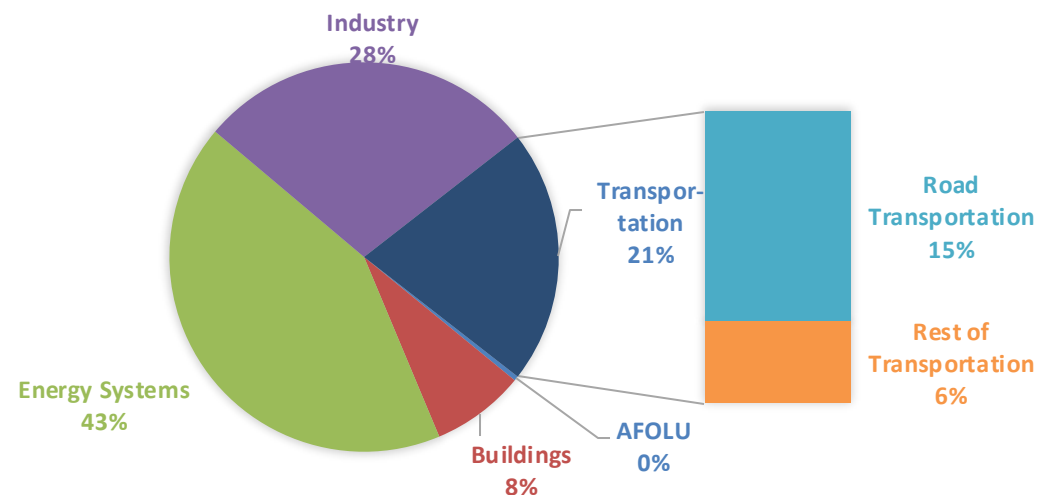
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

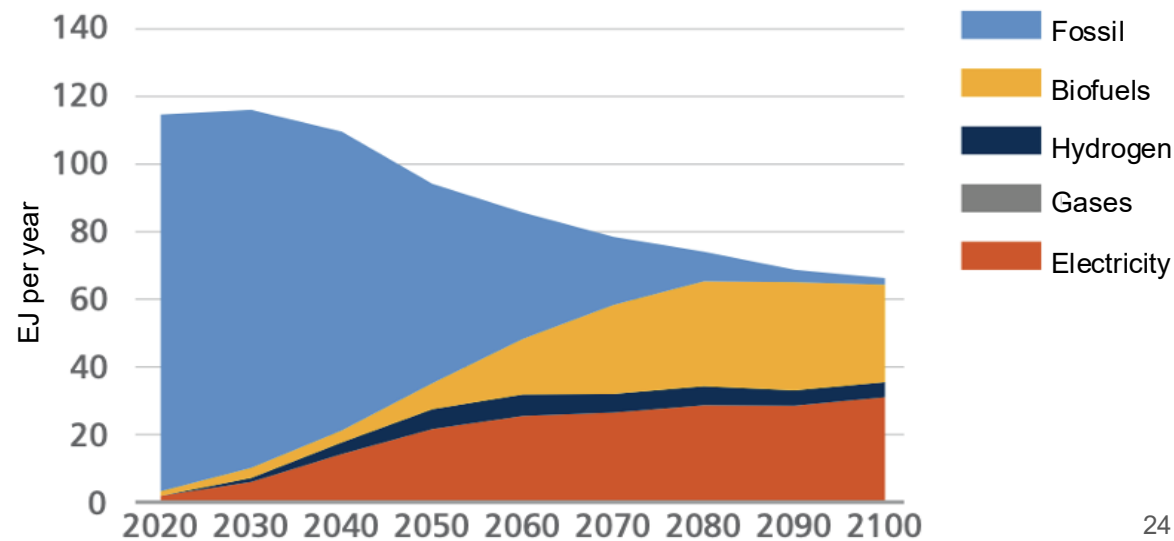
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

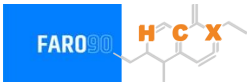
Current GHG Global Emissions Distribution



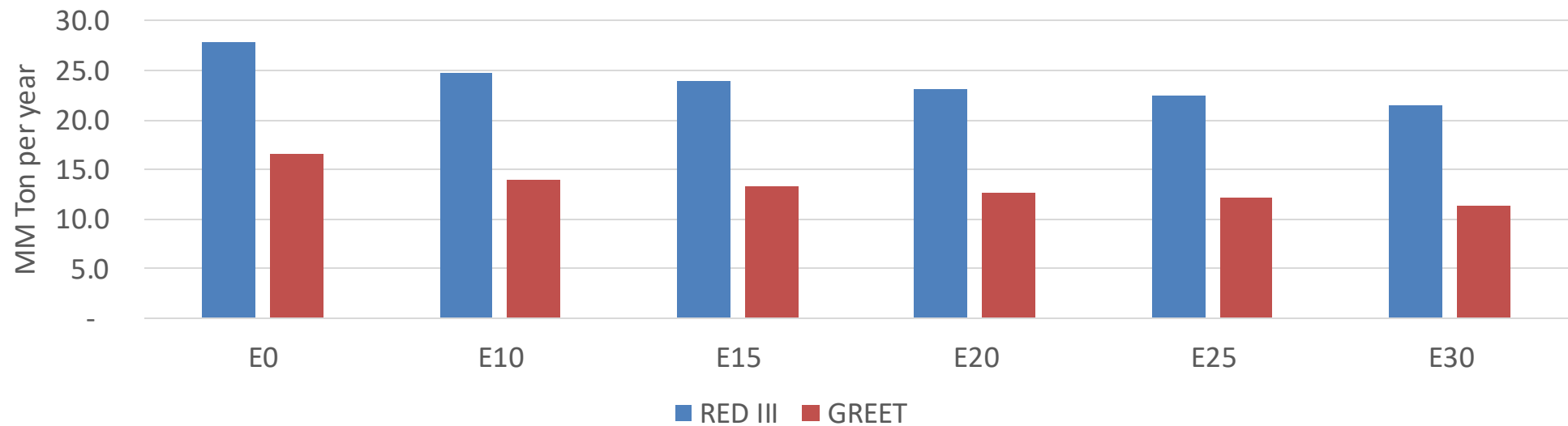
IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Ecuador – Gasoline Vehicle Fleet GEI Emissions



Life Cycle GHG Emission - Gasoline Vehicles



Country Emissions 2024 (MMT/year)	101.4
2035 Target (MMT/year)	71.0
Delta	30.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	16.0%	20.0%	23.6%	27.0%	31.3%
RED II/III	0.0%	11.1%	14.2%	16.9%	19.6%	22.7%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.7%	10.9%	12.9%	14.8%	17.1%
RED II/III	0.0%	10.1%	13.0%	15.5%	18.0%	20.8%